12/02/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

Regd. Office: Aakash Tower, 8,Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - 9

Answer Key

1.	(2)	37.	(3)	73.	(1)	109.	(3)	145.	(4)
2.	(2)	38.	(1)	74.	(2)	110.	(2)	146	(4)
3.	(4)	39.	(4)	75.	(2)	111.	(1)	147.	(4)
4.	(4)	40.	(4)	76.	(4)	112.	(2)	148.	(1)
5.	(4)	41.	(3)	77.	(4)	113.	(4)	149.	(4)
6.	(1)	42.	(4)	78.	(3)	114.	(4)	150.	(4)
7.	(2)	43.	(1)	79.	(4)	115.	(3)	151.	(3)
8.	(3)	44.	(2)	80.	(3)	116.	(1)	152.	(4)
9.	(2)	45.	(4)	81.	(2)	117.	(3)	153.	(1)
10.	(2)	46.	(4)	82.	(2)	118.	(1)	154.	(2)
11.	(2)	47.	(2)	83.	(1)	119.	(4)	155.	(3)
12.	(2)	48.	(2)	84.	(4)	120.	(3)	156.	(2)
13.	(3)	49.	(1)	85.	(1)	121.	(1)	157.	(4)
14.	(2)	50.	(2)	86.	(2)	122.	(4)	158.	(3)
15.	(1)	51.	(3)	87.	(3)	123.	(3)	159.	(1)
16.	(4)	52.	(4)	88.	(2)	124.	(3)	160.	(3)
17.	(2)	53.	(1)	89.	(3)	125.	(3)	161.	(3)
18.	(1)	54.	(1)	90.	(3)	126.	(2)	162.	(4)
19.	(4)	55.	(3)	91.	(2)	127.	(1)	163.	(4)
20.	(4)	56.	(3)	92.	(3)	128.	(2)	164.	(2)
21.	(3)	57.	(3)	93.	(1)	129.	(2)	165.	(2)
22.	(4)	58.	(3)	94.	(1)	130.	(4)	166.	(2)
23.	(2)	59.	(2)	95.	(2)	131.	(1)	167.	(3)
24.	(3)	60.	(1)	96.	(1)	132.	(3)	168.	(1)
25.	(4)	61.	(2)	97.	(4)	133.	(4)	169.	(4)
26.	(4)	62.	(2)	98.	(3)	134.	(3)	170.	(3)
27.	(1)	63.	(2)	99.	(3)	135.	(2)	171.	(2)
28.	(3)	64.	(2)	100.	(2)	136.	(1)	172.	(4)
29.	(1)	65.	(3)	101.	(4)	137.	(3)	173.	(4)
30.	(2)	66.	(3)	102.	(1)	138.	(2)	174.	(1)
31.	(1)	67.	(3)	103.	(1)	139.	(2)	175.	(3)
32.	(2)	68.	(1)	104.	(1)	140.	(4)	176.	(3)
33.	(3)	69.	(3)	105.	(3)	141.	(4)	177.	(2)
34.	(4)	70.	(1)	106.	(4)	142.	(2)	178.	(2)
35.	(3)	71.	(3)	107.	(2)	143.	(3)	179.	(1)
36.	(2)	72.	(4)	108.	(3)	144.	(3)	180.	(1)

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Test - 9

Hints and Solutions

PHYSICS

1. Answer (2)

When object is placed at the centre of sphere, the rays from the object fall normally on surface of sphere and emerge undeviated. So image form at *C* itself. Hence distance of image from surface is 6 cm.

2. Answer (2)

$$m = -\frac{v}{u},\ 2 = -\frac{v}{u} \therefore\ v = -2u$$

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}\ \therefore\ -\frac{1}{2u}+\frac{1}{u}=\frac{1}{-10}$$

So on solving $u = -5$ cm

3. Answer (4)

$$P_2 = P - P_1 = \frac{100}{80}-\frac{100}{20} = -3.75\,\text{D}$$

4. Answer (4)

Dispersive power

$\omega = \frac{\mu_v - \mu_R}{\mu - 1}$ by all four.

5. Answer (4)

$$\frac{1}{f}=\frac{\mu-1}{R}=\frac{1.5-1}{R}=\frac{1}{2R}$$

When placed in contact

$$\frac{1}{F}=\frac{1}{f}+\frac{1}{f}=\frac{1}{2R}+\frac{1}{2R}=\frac{1}{R}$$

6. Answer (1)

$$^{w}\mu_g = \frac{\mu_g}{\mu_w}=\frac{5/3}{4/3}=\frac{5}{4}$$

$$\sin C = \frac{1}{^{w}\mu_g}=\frac{1}{5/4}=\frac{4}{5}$$

$$\therefore\ C = \sin^{-1}\left(\frac{4}{5}\right)$$

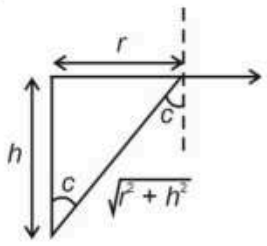
7. Answer (2)

$$i_1 = i_2 = \frac{3}{4}A \quad A + \delta = i_1 + i_2$$

$$\delta = i_1 + i_2 - A = \frac{3}{4}\text{A} + \frac{3}{4}A - A = \frac{A}{2}$$

$$\delta = \frac{60}{2} = 30^\circ$$

8. Answer (3)



$$\mu = \frac{1}{\sin C}=\frac{\sqrt{r^2+h^2}}{r}=\frac{5}{3}$$

$$25r^2 = 9(h^2 + r^2)$$

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$16r^2 = 9h^2$

$\therefore \quad r = \frac{3}{4}h = \frac{3}{4} \times 4 = 3\,\text{m}$

9. Answer (2)

For no deviation

$(\mu - 1)A = (\mu' - 1)A'$

$A' = \frac{(\mu - 1)A}{(\mu' - 1)} = \frac{(1.54 - 1)}{(1.92 - 1)} \times 5^\circ = 2.9^\circ$

10. Answer (2)

In shortsighted case concave lens of $f = -50$ cm should be used.

Hence, $P = \frac{100}{f} = -2\text{D}$

11. Answer (2)

$R.P. \propto \frac{1}{\lambda}$

$\frac{R_1}{R_2} = \frac{\lambda_2}{\lambda_1} = \frac{5000}{4000} = \frac{5}{4}$

12. Answer (2)

$\mu = \frac{\sin i}{\sin r} = \frac{\sin i}{\sin (i/2)}$

$= \frac{2\sin (i/2) \cdot \cos (i/2)}{\sin (i/2)} = 2\cos(i/2)$

$= \cos(i/2) = \frac{\mu}{2} = \frac{5/3}{2} = \frac{5}{6}$

$\Rightarrow (i/2) = \cos^{-1}\left(\frac{5}{6}\right)$

$= i = 2\cos^{-1}\left(\frac{5}{6}\right)$

13. Answer (3)

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, $u = -280$ cm, $v = ?$, $f = 20$ cm

$\frac{1}{-280} + \frac{1}{v} = \frac{1}{20} \quad \therefore \quad v = \frac{280}{15}\,\text{cm}$

and relative velocity of image v_i

$v_i = -\left(\frac{v}{u}\right)^2 \times \left(\frac{du}{dt}\right) = \frac{-(280/15)^2}{(-280)^2} \times 30$

$= \frac{-1}{15 \times 15} \times 30 = -\frac{2}{15}\,\text{m/s}$

Speed = $\frac{2}{15}\,\text{ms}^{-1}$

14. Answer (2)

$P_1 = 5\text{D}$, $P_2 = P - P_1 = 2 - 5 = -3\text{D}$

$\therefore \quad \frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2} = -\frac{P_2}{P_1} = \frac{3}{5}$

15. Answer (1)

$\theta = \frac{\ell}{r} = \frac{3}{D} \quad \therefore \quad D = \frac{3}{\theta}$

$\theta' = 1' = \frac{1}{60} \times \frac{\pi}{180}\,\text{rad}$

$D = \frac{3 \times 60 \times 180}{\pi}\,\text{m} = 10309\,\text{m} = 10\,\text{km}$

16. Answer (4)

$P = 2P_L + P_M$

$P_L = \frac{(\mu - 1) \times 2}{R}$, $R = 10$ cm

$\frac{1}{F} = \frac{2}{10} + \frac{2}{10} = \frac{2}{5}$

$-\frac{1}{F} = -\frac{2}{5}$

$F = -2.5$ cm

17. Answer (2)

For real image

$u = -u_1$, $v = -2u_1$, $f = -20$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\therefore \quad \frac{1}{-2u_1} + \left(\frac{1}{-u_1}\right) = \frac{1}{-20}$

$\therefore \quad \frac{-3}{2u_1} = -\frac{1}{20}$

$u_1 = 30$ cm

For virtual image

$u = -u_2$, $v = +2u_2$, $f = -20$ cm

$\frac{1}{2u_2} - \frac{1}{u_2} = \frac{1}{-20}$

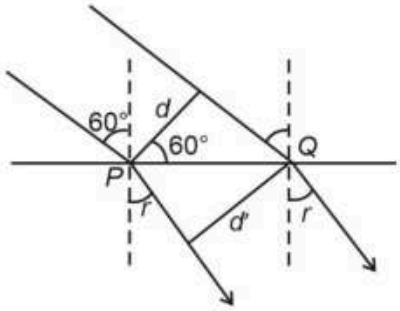
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$\therefore \quad u_2 = 10$ cm

Distance between object position

$= u_1 - u_2 = 20$ cm

18. Answer (1)



$d = PQ\cos 60°$

$d' = PQ\cos r = 2d\cos r$

$\frac{d'}{d} = \frac{\cos r}{\cos 60°} = \frac{\cos r}{1/2} = 2\cos r$

$\mu = \frac{\sin i}{\sin r}$

$\therefore \sin r = \frac{\sin i}{\mu} = \frac{\sin 60}{3/2} = \frac{\sqrt{3}/2}{3/2} = \frac{1}{\sqrt{3}}$

$\cos r = \sqrt{1-\sin^2 r} = \sqrt{1-\frac{1}{3}} = \sqrt{\frac{2}{3}}$

$\therefore \quad d' = 2\times 2\times\sqrt{\frac{2}{3}} = 3.26$ cm

19. Answer (4)

$M = \frac{f_0}{f_e} = 10 \quad \therefore \quad f_0 = 10\, f_e$

$10f_e + f_e = 110 \quad \therefore \quad f_e = 10$ cm, $f_0 = 100$ cm

When image is formed at least distance of distinct vision.

$M = f_0\left[\frac{1}{D} + \frac{1}{f_e}\right] = 100\left[\frac{1}{25} + \frac{1}{10}\right] = 14$

20. Answer (4)

When converging beam falls on concave lens the rays may converge, diverge or remain parallel.

21. Answer (3)

$f_0 = 1.5$ cm, $f_e = 2.5$ cm

$D = v_0 + f_e = 25$ cm, $\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$

$\frac{1}{22.5} - \frac{1}{u_0} = \frac{1}{1.5}$

$\therefore \quad -\frac{1}{u_0} = \frac{1}{1.5} - \frac{1}{22.5} = \frac{21}{1.5\times 22.5}$

$|v_0| = \frac{1.5\times 22.5}{21} \approx 1.6$ cm

For relaxed eye

$M = \frac{v_0}{u_0}\times\left(\frac{D}{f_e}\right) = \frac{22.5}{1.6}\times\frac{25}{2.5}$

$M = 140.6$

22. Answer (4)

The splitting of white light into component colours is called dispersion.

23. Answer (2)

$\delta = (\mu - 1)A$

or $A = \frac{\delta}{(\mu-1)} = \frac{4.8}{(1.62-1)} = \frac{4.8}{0.62} = 7.74°$

24. Answer (3)

$\frac{f_\ell}{f_a} = \frac{({}^a\mu_g - 1)}{({}^\ell\mu_g - 1)} = \frac{(1.5-1)}{\left(\frac{1.5}{4/3} - 1\right)} = 4$

$\Rightarrow \quad f_\ell = 4\times 20 = 80$ cm

25. Answer (4)

$m = \frac{I}{0} = \frac{f-v}{f}$

$\frac{I}{1.5} = \frac{25-75}{25} \Rightarrow I = -3$ cm

26. Answer (4)

$m = \frac{f_o}{f_e}$

27. Answer (1)

R.P. of telescope = $\frac{D}{1.22\,\lambda}$

It can be increased by increasing D and decreasing λ.

28. Answer (3)

To a fish inside water, bird appears to be at a larger distance than its actual distance.

29. Answer (1)

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\therefore \quad \frac{1}{v}-\frac{1}{f}+\frac{1}{u}=\frac{u+f}{uf} \therefore v=\frac{uf}{u+f}$$

$$v_1=\frac{-10f}{-10+f},\ v_2=\frac{-40f}{-40+f}$$

As images in same plane $v_2 = -v_1$

$$\frac{-40f}{-40+f}=\frac{10f}{-10+f}$$

$$\therefore \quad -400f+10f^2=400f-40f^2$$

$$50f^2-800f=0$$

$$\therefore \quad f=16 \text{ cm}$$

30. Answer (2)

$$\beta=\frac{\lambda' D'}{d'}=\frac{3\lambda\times 3D}{d/3}=\frac{27\,\lambda D}{d}$$

31. Answer (1)

$$\frac{I_1}{I_2}=\left(\frac{a}{b}\right)^2=\frac{9}{4} \ \therefore \ \frac{a}{b}=\frac{3}{2}$$

$$\frac{I_{max}}{I_{min}}=\frac{(a+b)^2}{(a-b)^2}=\frac{(3+2)^2}{(3-2)^2}=\frac{25}{1}$$

32. Answer (2)

$$\beta=\frac{\lambda D}{d}=\frac{6000\times10^{-10}\times 2}{4\times10^{-3}}=0.3\times10^{-3}\text{ m}$$

$= 0.3$ mm

33. Answer (3)

When unpolarised light passes through a polaroid its intensity becomes half.

34. Answer (4)

Doppler's shift when source reaching observer with changing speed.

35. Answer (3)

It is polarising angle by Brewster's law

$$\mu=\tan\theta_p=\tan(53^\circ)=\frac{4}{3}$$

36. Answer (2)

$$y'=\frac{2\lambda' D'}{b'}=\frac{600\times D\times b}{\frac{b}{2}\times 400\,D}=\frac{y'}{y}$$

$$y'=3y$$

37. Answer (3)

$$\theta=1.22\frac{\lambda}{D}=1.22\frac{5000\times10^{-10}}{10\times10^{-2}}=6.1\times10^{-6}\text{ rad.}$$

38. Answer (1)

$$n=\frac{(\mu-1)\,t}{\lambda}=\frac{(1.5-1)\times2\times10^{-6}}{5000\times10^{-10}}=2$$

39. Answer (4)

$$x=\frac{n\lambda D}{d}=(n+1)\frac{\lambda' D}{d}$$

$$\therefore \quad n\times 12000=(n+1)\times 1000$$

$$n=5$$

$$x=\frac{5\times(12000\times10^{-10})\times 2}{2\times10^{-3}}=6\times10^{-3}\text{ m}=6\text{ mm}$$

40. Answer (4)

Unpolarised intensity $= 100 = I_0$

Intensity from first Nicol prism $= \frac{I_0}{2}=50$

According to law of Malus

$$I=I_0'\cos^2\theta=50\cos^2(60^\circ)$$

$= 50 \times (\ell/2)^2 = 12.5$

41. Answer (3)

$$\frac{\Delta\lambda}{\lambda}=\frac{v}{C} \ \therefore \ \Delta\lambda=\frac{v}{C}\times\lambda$$

$$=\frac{10^5}{3\times10^8}\times5700=1.9\text{ Å}$$

42. Answer (4)

$$10\lambda'\frac{D}{d}=(2\times6-1)\frac{\lambda}{2}\frac{D}{d}=5.5\frac{\lambda D}{d}$$

$$\frac{\lambda}{\lambda'}=\frac{10}{5.5}=1.8 \ \therefore \ \mu=\frac{\lambda}{\lambda'}=1.8$$

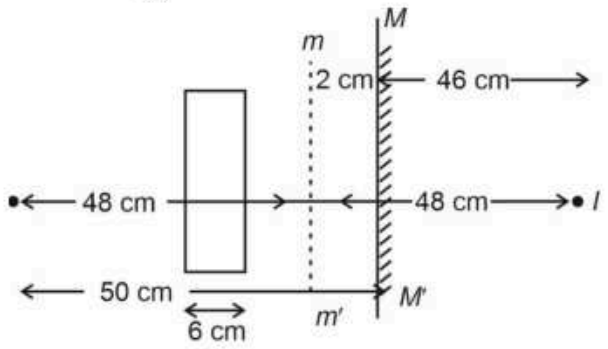
43. Answer (1)

$p = mv$, by wave equation

$$\lambda=\frac{h}{P}=\frac{h}{mv}\ \text{decreases}$$

$$\beta=\frac{\lambda D}{d}\ \text{will decrease}$$

44. Answer (2)



Shift due to slab = $6\left[1-\frac{2}{3}\right]=2$ cm

So glass shift mirror from M' to mm' distance of object from this virtual mirror will be

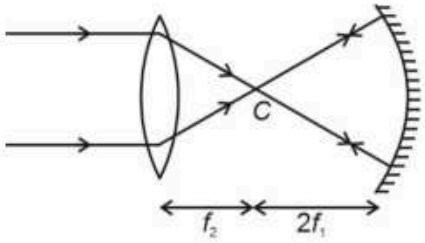
$= 50 - x = 50 - 2 = 48$ cm

The virtual mirror will form image of object O at a distance 48 cm behind it, so distance of image from actual mirror MM' is $= 48 - 2 = 46$ cm

Distance of image as seen by observer

$= 50 + 46 = 96$ cm

45. Answer (4)

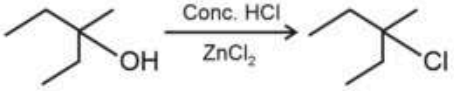


$d = 2f_1 + f_2$

CHEMISTRY

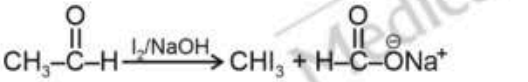
46. Answer (4)

Tertiary alcohol on reaction with Lucas reagent (Conc. HCl and $ZnCl_2$) gives alkyl halide immediately which is the cause of turbidity.

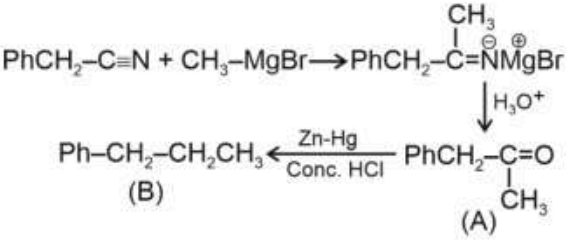


47. Answer (2)

Compounds containing $CH_3-\overset{O}{\overset{\|}{C}}-$ group or $CH_3-\underset{OH}{\underset{|}{CH}}-$ group give positive iodoform test.



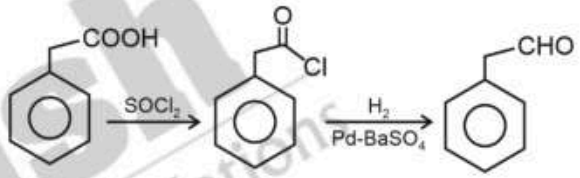
48. Answer (2)



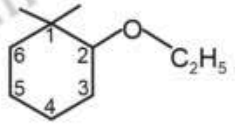
49. Answer (1)

In Etard reaction, toluene is oxidised to benzaldehyde. Here no C-C bond formation takes place.

50. Answer (2)

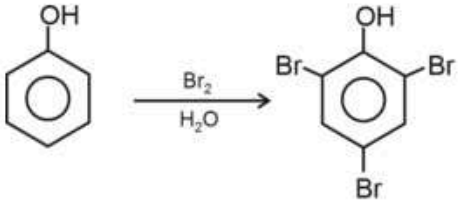


51. Answer (3)



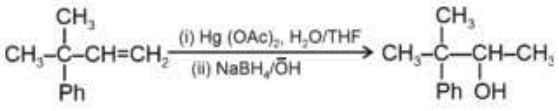
2-Ethoxy-1,1-dimethylcyclohexane

52. Answer (4)



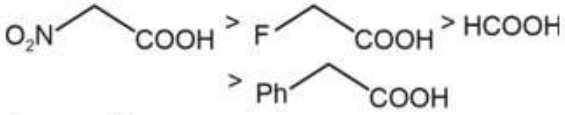
53. Answer (1)

In oxymercuration-demercuration reaction, OH is added to double bond containing less number of hydrogen atoms. Rearrangement does not occur in this reaction.



54. Answer (1)

Correct order of acidic strength is



55. Answer (3)

Compound which is stronger acid than carbonic acid will liberate CO_2 on reaction with $NaHCO_3$.

56. Answer (3)

Aromatic aldehyde does not respond to Fehling's test.

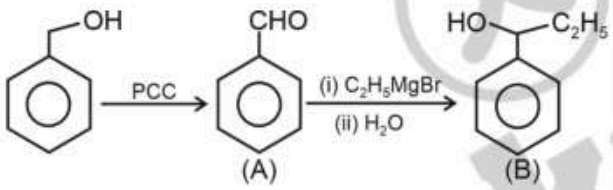
57. Answer (3)

Ketones react with one equivalent of monohydric alcohol in presence of dry HCl to yield alkoxyalcohols, known as hemiketals.

58. Answer (3)

Aliphatic aldehydes react fastest with HCN as carbonyl carbon of aliphatic aldehyde is highly electrophilic in nature and tetrahedral intermediate formed is less hindered.

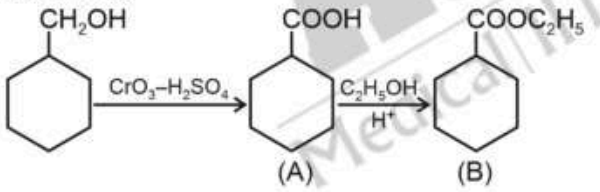
59. Answer (2)



60. Answer (1)

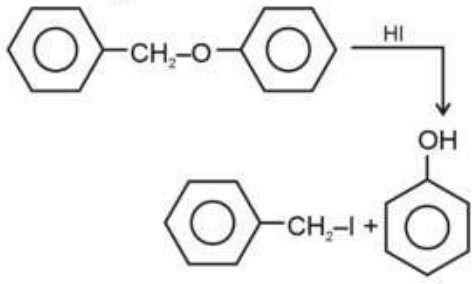
Secondary alcohol is oxidised to ketone by copper at 573 K.

61. Answer (2)



B is an ester.

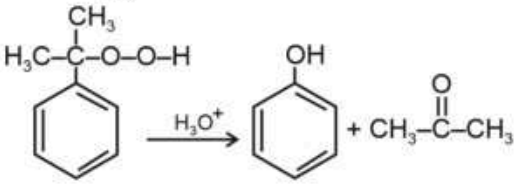
62. Answer (2)



63. Answer (2)

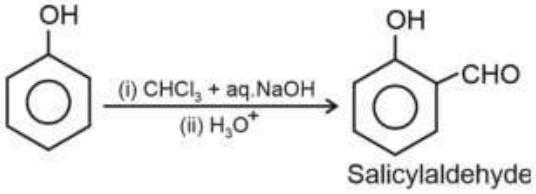
Increasing number of electron releasing group ($-CH_3$) increases electron density on oxygen leading to decrease in polarity of O–H bond.

64. Answer (2)



65. Answer (3)

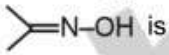
Reimer-Tiemann reaction



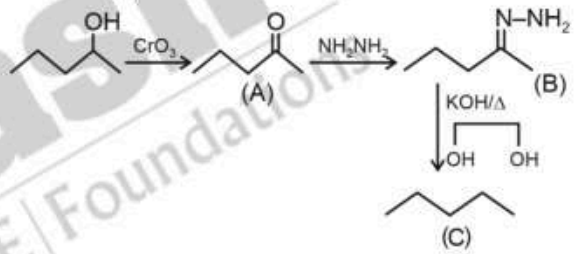
66. Answer (3)

Ether does not react with sodium metal.

67. Answer (3)

 is called oxime.

68. Answer (1)

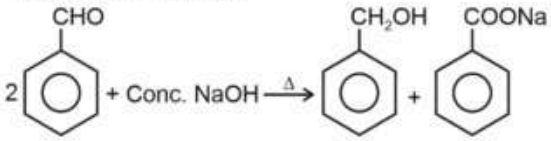


69. Answer (3)

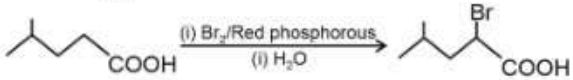
For aldol condensation to occur the carbonyl compound should contain at least two alpha hydrogens.

70. Answer (1)

Cannizzaro reaction



71. Answer (3)



Hell-Volhard-Zelinsky reaction.

72. Answer (4)

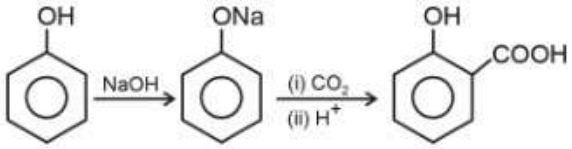
The carbonyl compounds which do not contain enolizable hydrogen will not give keto-enol tautomerism.

73. Answer (1)

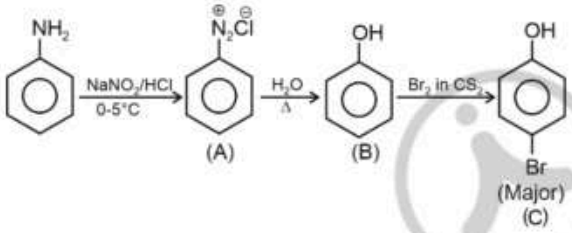
In allyl alcohol the –OH group is attached to a sp^3 hybridised carbon next to carbon-carbon double bond.

74. Answer (2)

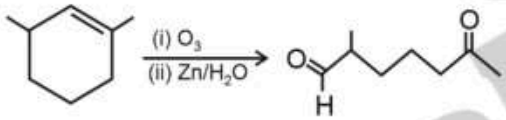
Kolbe's reaction



75. Answer (2)



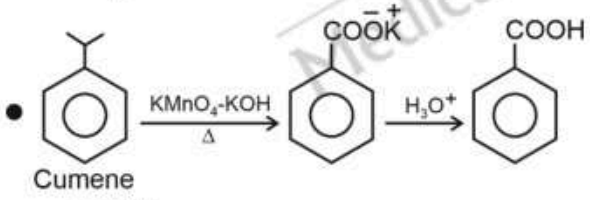
76. Answer (4)

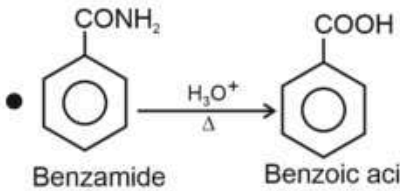


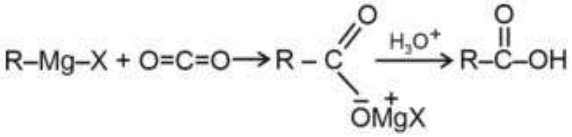
77. Answer (4)

Electron withdrawing group present at para position makes carbonyl carbon more electrophilic hence rate of hydrolysis increases.

78. Answer (3)





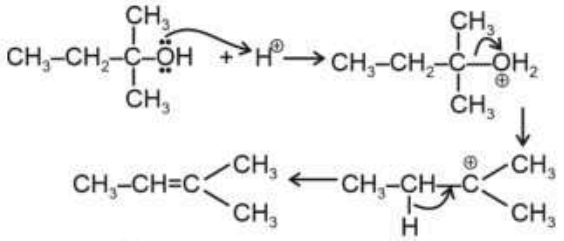


79. Answer (4)

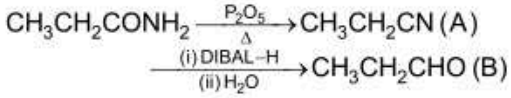
The commercial ethanol is made unfit for drinking by mixing in it copper sulphate and pyridine. It is known as denaturation of alcohol.

80. Answer (3)

Tertiary alcohol is dehydrated most easily in acidic medium via formation of carbocation.



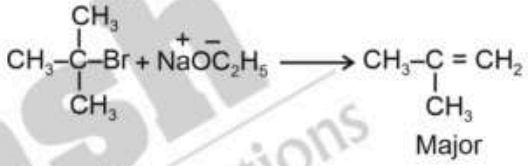
81. Answer (2)



82. Answer (2)

Primary alcohol gives red colour in Victor Meyer's test.

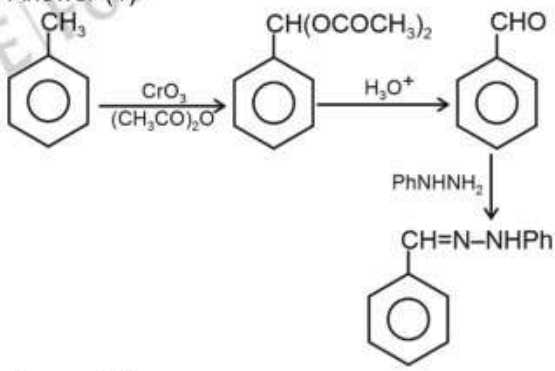
83. Answer (1)



84. Answer (4)

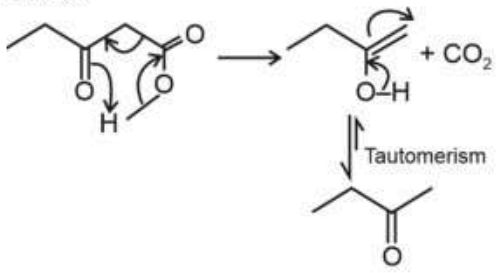
Ketones do not react with Tollen's reagent.

85. Answer (1)



86. Answer (2)

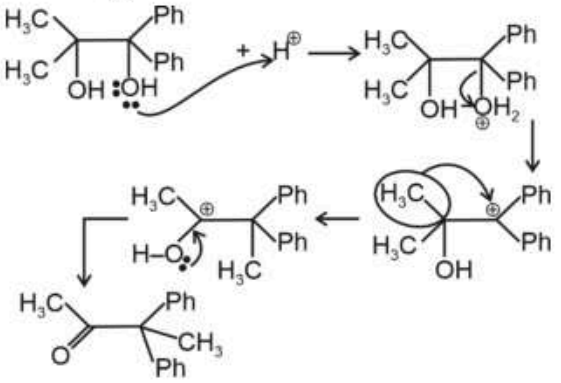
β-keto acid decarboxylates most easily on heating.



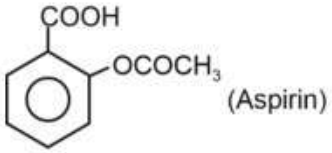
87. Answer (3)

Boiling point of ethanol (351 K) is higher than acetone (329 K) because of intermolecular H-bonding in ethanol.

88. Answer (2)

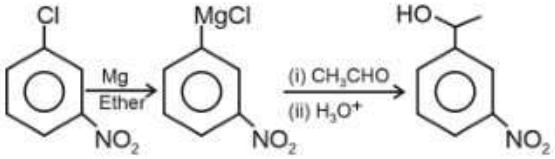


89. Answer (3)



(Aspirin)

90. Answer (3)



(A) (B)

BOTANY

91. Answer (2)

Viruses multiply only inside the cells.

92. Answer (3)

Curd is more nutritious than milk as it contains a number of vitamins especially B_{12}. Lactic acid bacteria present in curd play very beneficial role in checking disease causing microbes.

93. Answer (1)

Lactobacillus converts milk into curd. It is a heterotrophic bacterium.

94. Answer (1)

Cheese is one of the oldest food items in which microbes are used.

95. Answer (2)

Swiss cheese is ripened with the help of *Propionibacterium sharmanii* and Roque-fort cheese is ripened by fungus *Penicillium roqueforti*.

96. Answer (1)

Bread is prepared from dough, which is fermented using Baker's yeast *Saccharomyces cerevisiae*.

97. Answer (4)

Puffed-up appearance of dough is due to production of CO_2 during fermentation.

98. Answer (3)

Toddy is traditional drink of some parts of India made by fermenting sap from palm.

99. Answer (3)

Beer is prepared without the process of distillation but brandy is prepared by distillation. So, beer has less percentage of alcohol than brandy.

100. Answer (2)

Antibiotics means against life, in context of disease causing organisms, whereas for human beings, it is prolife.

101. Answer (4)

The first antibiotic that was discovered by Alexander Fleming is penicillin.

102. Answer (1)

Full potential of penicillin as an effective antibiotic was established by Ernst Chain and Howard Florey.

103. Answer (1)

While working on *Staphylococci*, Alexander Fleming observed that the presence of a mould *i.e.*, *Penicillium notatum* inhibits the growth of this bacterium.

104. Answer (1)

Clostridium butylicum is used for commercial production of butyric acid.

105. Answer (3)

Lipases are used in detergent formulation and are helpful in removing oil stains from laundry.

106. Answer (4)

Cyclosporin A is produced by the fungus *Trichoderma polysporum* and statins by yeast-*Monascus perpureus*. They belong to different classes of fungi. Both of them reproduce asexually.

107. Answer (2)

Streptokinase is produced by a bacterium *Streptococcus*. It is a blood clot buster that removes clots from blood vessels of patients who have undergone myocardial intraction.

108. Answer (3)

The effluent from the primary settling tank is taken for secondary treatment.

109. Answer (3)

In sewage treatment, secondary treatment is also known as biological treatment. This process is done by both aerobic as well as anaerobic microorganisms.

110. Answer (2)

In secondary treatment, sewage water is treated till the BOD is reduced significantly.

111. Answer (1)

In aerobic sludge digester, anaerobic methanogenic bacteria digest other bacteria and fungi present in sludge and produce gases such as methane, hydrogen sulphide and carbon dioxide.

112. Answer (2)

In biogas, the major component is of methane which is about 50-70%.

113. Answer (4)

Organic farming works to create a system where the pests are not eradicated completely but kept at manageable levels.

114. Answer (4)

Baculoviruses are pathogens that attack insects and other arthropods.

115. Answer (3)

Integrated pest management programme does not include complete destruction of any plant species.

116. Answer (1)

Nostoc, *Oscillatoria* and *Anabaena* are photosynthetic nitrogen fixing organisms.

117. Answer (3)

A population is group of same species inhabiting an area.

118. Answer (1)

Ecology is the study of interactions among organisms and between the organisms and their physical environment.

119. Answer (4)

Tropical forests receive maximum mean annual rainfall.

120. Answer (3)

Topographic factors are physical factors related to slope, altitude and others concerned with surface behaviour of earth.

121. Answer (1)

Salinity of inland water is less than 5.

122. Answer (4)

Individuals of same species in a population occupy the same niche.

123. Answer (3)

Mammals such as polar bears are also stenothermal.

124. Answer (3)

Global warming push tropics into temperate areas and temperate areas towards pole. This is shifting of climate zones.

125. Answer (3)

Animals use variation in light intensities and photoperiod as cues for timing the foraging, reproductive and migratory activities.

126. Answer (2)

Regulators are able to maintain homeostasis by physiological as well as behavioural means and have constant body temperature.

127. Answer (1)

Small animals have large surface area relative to their volume. Therefore, they tend to lose body heat very fast when it is cold outside.

128. Answer (2)

Migration is escape in space.

129. Answer (2)

Decreasing the binding affinity of haemoglobin is one of the ways to compensate low oxygen availability at higher altitudes.

130. Answer (4)

Let the initial population of fishes be N_0.

Therefore, after one year the number of fishes,

$N_1 = N_0 + (300 - 120)$

$780 = N_0 + 180$

$N_0 = 780 - 180 = 600$

Birth rate = $\frac{300}{600}$ = 0.5 per fish per year

131. Answer (1)

A population with bell-shaped age pyramid is said to be stable as there is no increase or decrease in the size of population.

132. Answer (3)

Occurrence of CAM pathway for photosynthesis is a physiological adaption.

133. Answer (4)

Plants offer juicy and nutritious fruits in the form of rewards for their seed dispersers.

134. Answer (3)

Commensalism is a '+' '0' type of interaction.

135. Answer (2)

Amensalism is an interaction between two organisms of different species in which one species inhibits the growth of other species. *Penicillium* inhibits the growth of bacteria.

ZOOLOGY

136. Answer (1)

Bone marrow and thymus gland are considered as primary lymphoid organs. Other given organs are included in secondary lymphoid organs.

137. Answer (3)

Ascariasis is caused by infestation of embryonated eggs of *Ascaris* which present in soil, through faeco-oral route.

138. Answer (2)

Plasmodium vivax is a causative agent of benign malaria.

139. Answer (2)

HIV is a member of a group of viruses called retrovirus which have an envelope enclosing the RNA genome.

140. Answer (4)

'NACO' abbreviation is used for "National AIDS Control Organisation".

141. Answer (4)

All types of barriers are part of innate immunity but memory of first encounter is stored in acquired immunity.

142. Answer (2)

Vaccination provides artificial active immunity to children. Both, IgA antibodies present in colostrum and IgG antibodies that cross the placenta provide natural passive immunity to new born babies and foetus respectively.

143. Answer (3)

Variable regions of both light and heavy chain are present at their N-terminals and provide binding site for antigen.

144. Answer (3)

Physical barriers	–	Skin on our body
Physiological barriers	–	Acid in the stomach
Cellular barriers	–	NK cells
Cytokine barriers	–	Interferons

145. Answer (4)

Frequent vomiting is a characteristic of cholera which can lead to alkalosis.

146 Answer (4)

Slaughtering of livestock is not included under the objective of animal husbandry.

147. Answer (4)

Improving growth rate of livestock is also an aim of animal breeding.

148. Answer (1)

Milk yield is primarily dependent on the quality of breeds in the dairy farm.

149. Answer (4)

Continued inbreeding, especially close inbreeding usually reduces fertility and even productivity. This is called inbreeding depression.

150. Answer (4)

Mule is a product of crossing between male donkey and female horse. Such type of breeding is called interspecific hybridisation.

151. Answer (3)

Houseflies act as mechanical carrier and serve to transmit the parasite from faeces of infected person to food and food products, thereby contaminating them.

152. Answer (4)

Gametogenesis in *Plasmodium* starts within mosquito's gut.

153. Answer (1)

Bacterial infections such as *Streptococcus pneumoniae* and *Haemophilus influenzae* are responsible for the disease pneumonia.

154. Answer (2)

Primary and secondary immune responses are carried out with the help of both B and T-lymphocytes. Intensified response with the same kind of pathogen elicits anamnestic response.

155. Answer (3)

Cell-mediated immune response performed by T-lymphocytes is responsible for graft-rejection or rejection of transplanted organ.

156. Answer (2)

HIV contains two single stranded RNA. HIV infected mother can transmit HIV but cannot transmit AIDS.

157. Answer (4)

HIV is not transmitted by shaking hands and sharing the food.

158. Answer (3)

The bone marrow is main lymphoid organ where all blood cells including lymphocytes are produced.

159. Answer (1)

IgG is most predominant antibodies of secondary response.

160. Answer (3)

It is estimated that more than 70% of the world livestock population is in India and China but they contribute only 25% of farm production.

161. Answer (3)

Pisciculture – Rearing of fishes

Sericulture – Rearing of silk moth

Aquaculture – Rearing of aquatic plants and animals.

162. Answer (4)

Bird flu is caused by a virus.

163. Answer (4)

Pollination in wheat occurs through wind.

164. Answer (2)

Proper food and water is required for poultry farm management. The temperature of poultry farm should be optimum.

165. Answer (2)

Cancer cells are more rapidly dividing cells due to loss of control over the cell division. So, they are easily damaged than normal cells due to loss of regulatory machinery for cell division.

166. Answer (2)

Only morphine is used as analgesic. Marijuana belongs to category hallucinogen.

167. Answer (3)

Codeine – *Papaver somniferum*

Charas – *Cannabis sativa*

Cocaine – *Erythroxylum coca*

LSD – *Claviceps purpurea*

168. Answer (1)

Hallucinogens are produced by plants such as Cannabis sativa, Datura species, Atropa belladona and fungus Claviceps purpurea.

169. Answer (4)

Emphysema and lung carcinoma are very common diseases associated with cigarette smoking.

170. Answer (3)

Coca alkaloid/cocaine interferes with the transport of inhibitory neurotransmitter dopamine in certain part of brain and stimulates CNS.

171. Answer (2)

Excessive dose of drugs may lead to coma or death due to respiratory failure.

172. Answer (4)

Spreading of cancer cells from one place to another in the body where they are lodged to grow into new tumor is called metastasis.

173. Answer (4)

Histamine and serotonin are mainly responsible for allergic reactions and are released from mast cells.

174. Answer (1)

The side effects of the use of anabolic steroids included agressiveness and mood swings in both male and female. But breast enlargement is side effect only in males.

175. Answer (3)

Smoking is associated with increase incidence of lung cancer, urinary bladder cancer and throat cancer.

176. Answer (3)

Breeding between same breed is called inbreeding, while crosses between different breeds are called outbreeding.

177. Answer (2)

Catla, Rohu, Common carps – Edible fresh water fishes

Hilsa, sardine, mackerel and pomfrets – Edible marine fishes.

178. Answer (2)

Insemination of genetic mother is done after superovulation. When fertilized eggs are transferred to surrogate mother, genetic mother becomes free for another round of superovulation.

179. Answer (1)

Blue revolution – Fishery industry

White revolution – Dairy farms

Green revolution – Agriculture

180. Answer (1)

Viral RNA is introduced into cell and after reverse transcription, viral DNA incorporates into host genome

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